

Class 1.7 — Capsule Build & First Splash

Sunday, July 26, 2026 · 3 hours · Mr. Holcombe's house

THE ONE GOAL

Every kid in the water, comfortable; one capsule successfully built and dropped; control-box smoke test clean. The Phase 1 exit gate has been moved forward in three meaningful ways.

BEFORE YOU TEACH THIS

- Watch the MATE "General ROV Safety" presentation end-to-end the night before. Assign the kids to watch it Friday or Saturday.
- Build one capsule yourself the night before. Time it. Mix one batch of epoxy so you remember the working-time pressure.
- Walk the pool deck without kids the day before if at all possible. Note the deepest point, the changing room route, the lifeguard chair location, the first-aid kit location.
- Read pages 252–267 (§6.1–6.2 — Buoyancy/Ballast tech notes) and page 765 (Safety index) of the Underwater Robotics textbook.
- Confirm the lifeguard contract is in writing. Confirm photo releases are on file for every kid expected in frame. These are HARD GATES — no exception, no judgment call, no "we'll get it after."
- Internalize the parent lane for today specifically: the smoke test is the kids', the capsule drop is the kids', the towel inspection is the kids'. You photograph. You do not power-up. You do not retrieve. You do not declare pass/fail. They do all three.

WHY THIS CLASS MATTERS

1.7 is the day Phase 1 stops being theoretical. The control box gets powered for the first time, the team gets in the water for the first time, and the first dry-dive count starts. Three Phase-1 exit-gate beats land in a single Sunday.

The operational complexity is the highest of any meeting so far — two safety briefs, a travel block, a smoke test, a pool entry, and a documented dive log. Do every block in order; resist the temptation to combine the safety briefs into one. The workshop brief and the poolside brief are different briefs and they live on different decks for a reason.

HOW THE MEETING RUNS — BLOCK BY BLOCK

0:00 – 0:10 · Workshop-half safety brief

Tighter than the usual brief — we have ten minutes, not fifteen. The four standing questions, then the workshop-half card with epoxy and PVC additions. Reinforce: ventilation on before epoxy comes out, gloves at the glue station, dry-eye protection for sanding.

Assign the four rotating roles. The Photographer is in for a heavy day; warn them now.

0:10 – 1:10 · Capsule build

Sixty minutes hands on. The constraint is marine-epoxy working time — about 5 minutes after mix. Mix in two batches, one for each cap, so we never have a kid holding a setting cap that hasn't gone on yet.

Towel goes in BEFORE the second cap. That's the gotcha that the SeaMATE Guide flags and that first-time builders miss. Coach John (or whoever holds the bench) calls it out before the second cap is mixed.

1:10 – 1:30 · Travel to the pool

Capsules go in a padded box, end-caps up — the epoxy needs another 30+ minutes to fully cure and the trip is the cure time. Battery in its enclosure, DISCONNECTED, with two-person verify before the car door closes.

Confirm at the deck: who is driving home which kid. Parents need to know the pickup is at the pool, not the workshop.

1:30 – 1:45 · Poolside safety brief

Different brief from the workshop. Run the eight standing rules on the slide, on the deck, with the kids standing. Introduce the lifeguard by name; let the lifeguard say one thing if they're willing.

Assign buddies before any kid touches the water. Write the buddy pairs in the engineering notebook so we can reconstruct who was where if there's ever an incident report.

1:45 – 2:15 · Every kid in the water — comfort first

Thirty minutes of pool, no agenda. No ROV, no tether, no mission. The kids swim around, get used to the depth, get used to the deck, get used to operating as a team in the water. This is the Phase 1 exit-gate beat we don't compromise.

Advisor stays on the deck — buddies tracked, lifeguard within line of sight, hydration available. Phone the parent of any kid who needs to be pulled out for any reason; cold, earache, dizziness, anything.

2:15 – 2:30 · Smoke test of the control box

ON THE DECK, not in the water. The smoke test is the moment the entire 1.5–1.6 discipline cashes in. Read the script on the slide; do not deviate. Polarity verified by two people. Watt meter shows expected idle current. If anything is off — smoke, smell, audible pop, current spike — disconnect immediately, document, debrief.

If the smoke test is clean, document the idle current in the engineering notebook and move to capsule drop #1. If it isn't, we don't drop today; we troubleshoot back at the workshop and re-test next week.

2:30 – 2:50 · Capsule drop #1 + towel check

Underhand toss to the deepest point. Wait sixty seconds. A buddy-paired kid retrieves. Dry the outside before opening. The towel decides — bone-dry passes, even a damp spot fails.

The kid who built the capsule opens it. The kid who ran the meter-check at 1.6 inspects the towel. The Notetaker logs the result with a photo of the towel. Date, time, pass/fail. That log is the proof at the regional that the dry-dive gate was hit.

2:50 – 3:00 · Wrap-up at the deck

Brief. Read the three student items, confirm owners. Thank the lifeguard out loud, in person — not a text after. Thank the host out loud, in person. The kids should see that.

Tool Steward leads the deck pack-out. Battery confirmed DISCONNECTED before it goes in the car. The closer can run at the deck or back at the workshop — either works.

THE PARENT LANE — SECTION 4.3

Today's discipline call: every consequential moment is the kids'. The smoke test runs from the kid who did the final meter-check at 1.6; the capsule drop is built and dropped and inspected by kids; the towel inspection's verdict comes from a kid. You photograph the moments, you do not declare them. The judges in April will ask "who did the smoke test?" — the team that hears a kid's name wins the point. The team that hears "the coach did it" does not.

COMMON PROBLEMS & QUICK FIXES

Symptom	Quick fix
Epoxy cap set before the kid was ready to attach it	Discard it. Mix a fresh batch. Don't try to re-soften — partially cured epoxy never re-bonds reliably.
O-ring twists when the cap is seated	Pull the cap off, re-seat the O-ring slowly, lubricate again with silicone grease, re-seat the cap.
Capsule floats when it should sink (or vice versa)	Add a small amount of ballast (lead shot or steel washers) for sinkers; add a small foam plug for floaters. Note the ballast in the dive log; we
Capsule retrieves with a damp towel after 60 seconds	Fail. Open both end caps, dry the interior, inspect the O-rings for nicks or twists, re-epoxy if the cap-to-pipe joint shows water trace. Re-test
Smoke test reading shows current spike or unexpected draw	Disconnect immediately. Visually inspect the board — look for any joint that looks different than it did at 1.6 sign-off. Re-meter every pad. Do
Kid is in physical distress during the swim block	Lifeguard FIRST! Buddy stays with the kid! Get the parent on the phone immediately. Do not minimize, do not negotiate — out of the water, parent contacted, decision made together.

WHAT "DONE" LOOKS LIKE

- Every kid was in the water and stayed for the full thirty-minute comfort block.
- Control box smoke test produced expected idle current with no anomalies; result signed off in the engineering notebook.
- Capsule drop #1 happened. Pass or fail, the result is logged with a photo of the towel.
- Lifeguard and pool host both thanked in person before the team left the deck.
- Battery returned to the workshop DISCONNECTED with two-person verify.
- Photos uploaded to the team folder tonight, while the colors are still fresh.

IF YOU FALL BEHIND

Three protect-the-priority rules tonight. (1) If the capsule build runs long, cut the swim block to twenty minutes — but DO NOT skip the every-kid-wet beat. The exit gate depends on it. (2) If the smoke test surfaces an anomaly, do NOT drop the capsule. The capsule drop without a working board has no diagnostic value; the troubleshoot block at next week's workshop has more. (3) If the lifeguard pulls the team from the water for any reason, the pull stands. We re-schedule, we don't argue.